

Pharmacotherapy of Type 2 Diabetes

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Learning Objectives

- Describe the indications, mechanisms of action, distinctive and adverse effects of medications used to treat type 2 diabetes
- Identify appropriate initial and add-on therapy for a patient with type 2 diabetes
- Identify clinically-important interactions involving anti-hyperglycemic medications
- Discuss general issues to consider when choosing a specific medication

Drug	Efficacy	Hypoglycemia	weight	Cardiovascular	cost	Clinical considerations
Metformin	High	none	Mild	Neutral	Low	Diarrhea B12 def Avoid when ill(lactic acidosis)
SGLT2 inhibitors	High to medium	none	Mild to moderate	Yes Renal protection	Medium	Genital yeast infection Fournier gangrene Stop when ill Euglycemia DKA
GLP-1 agonist	High	none	Very High	Yes	High	Risk of medullary thyroid cancer in rats' Cholecystitis/pancreatitis GI side effect Stop before any surgery x 1 week
DUAL GLP1 ,GIP agonist	High	none	Very High	Yes	High	As above

Drug	Efficacy	Hypoglycemia	weight	Cardiovascular benefit	cost	Clinical considerations
DPP4 inhibitor	Moderate	none	Neutral	Neutral	Medium	Pancreatitis Rare- bullous pemphigoid
Sulfonylurea	High	High	Gain	Neutral	Low	Careful in chronic kidney disease
Thiazolidinediones	High	none	Gain	Increase risk of heart failure	Medium	Benefit in Fatty liver
Insulin	High	High	Gain	Neutral	Medium	

Questions?

Applications

Case 1

A 45-year-old white man comes to the office for follow-up of type 2 diabetes. He has no acute complaints. The patient has taken metformin since being diagnosed with diabetes 6 years ago, after efforts at lifestyle change produced only slight improvements in blood sugar. He had some initial gastrointestinal side-effects that improved with time, and otherwise tolerated the medication and dose increases without problems. The patient lost several pounds with metformin and lifestyle changes but has regained some of that weight in the years since. His blood glucose was initially well-controlled, but for the past year his hemoglobin A1c has been above the therapeutic goal and the dose of metformin was increased.

Along with type 2 diabetes, the patient has history of hyperlipidemia. His current medications are metformin 1000 mg twice daily (last increased 3 months ago), pravastatin 10 mg daily, a multivitamin, and occasional ibuprofen for musculoskeletal aches. He has family history of type 2 diabetes in his mother and brother. He does not smoke. He is very apprehensive about taking “shots” or pills that could damage his pancreas.

The patient is 175 cm (5'9”) tall and weighs 92 kg (202 lbs), with a body mass index of 30. His blood pressure is 128/82, sensation in his extremities is intact, and there are no other abnormalities on physical exam. Recent fasting blood tests showed glucose 140 mg/dL, hemoglobin A1c 8.3%, total cholesterol 194 mg/dL, LDL 120, HDL 42, and triglycerides 160.

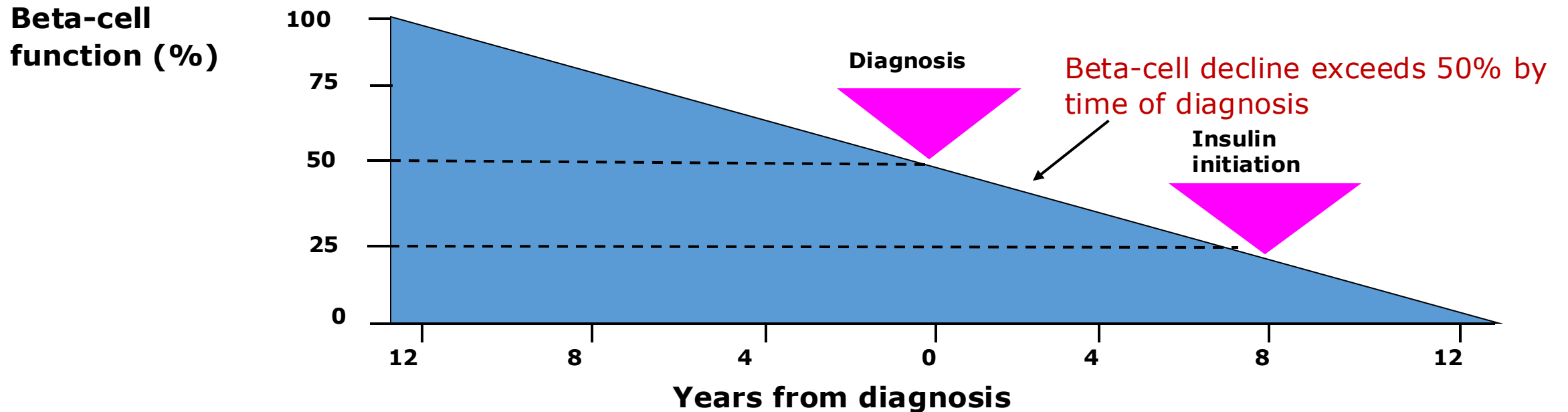
Question 1

Which of the following processes is probably most responsible for the increase in this individual's blood glucose concentrations over the past year?

- A. decreased insulin secretion from islet cells
- B. decreased insulin sensitivity of tissues
- C. down-regulation of AMPK in response to metformin

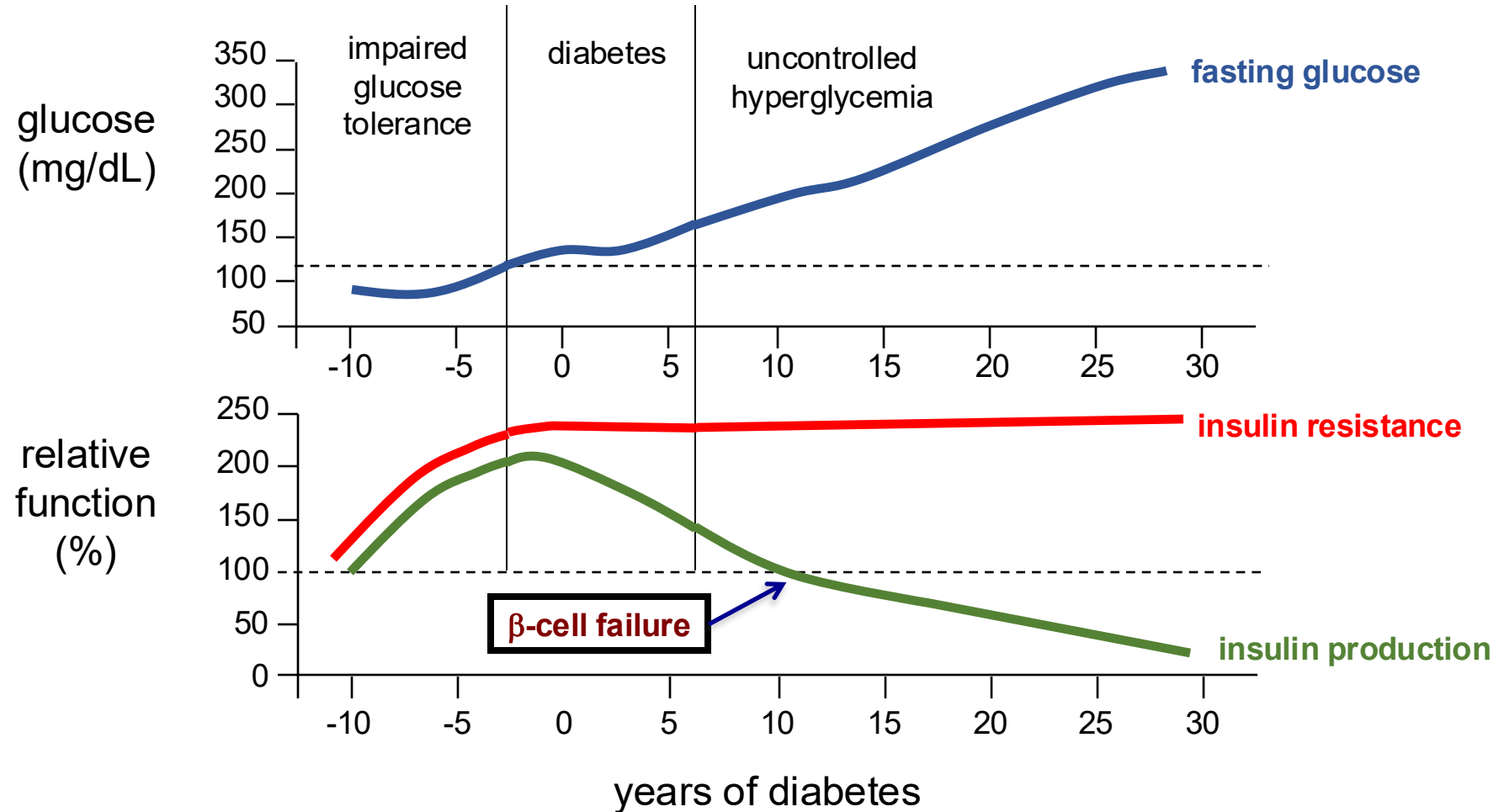
Beta-cell function declines as diabetes progresses (Type 2)

Beta-cell function decline over time



Adapted from Lebovitz H. Diabetes Rev 1999;7:139-153.
2004 Roper Survey of the US Diabetes Market,
courtesy of GfK Market Measures.

Progression of type 2 diabetes



Question 2

After confirming that the patient is following lifestyle recommendations and taking current medications as prescribed, which of the following is the best next step in treating his type 2 diabetes?

- A. add a medication
- B. allow more time at the current dose of metformin
- C. increase the dose of metformin
- D. replace metformin with another non-insulin medication
- E. stop one of the patient's other medications

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Question 3

The physician and patient decide to start an additional medication. Which of the following classes of medication would be the best choice?

- A. α -glucosidase inhibitor
- B. DPP-4 inhibitor
- C. Thiazolidinedione
- D. SGLT2 inhibitor
- E. sulfonylurea

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Use of glucose lowering medications in management of Type 2 diabetes

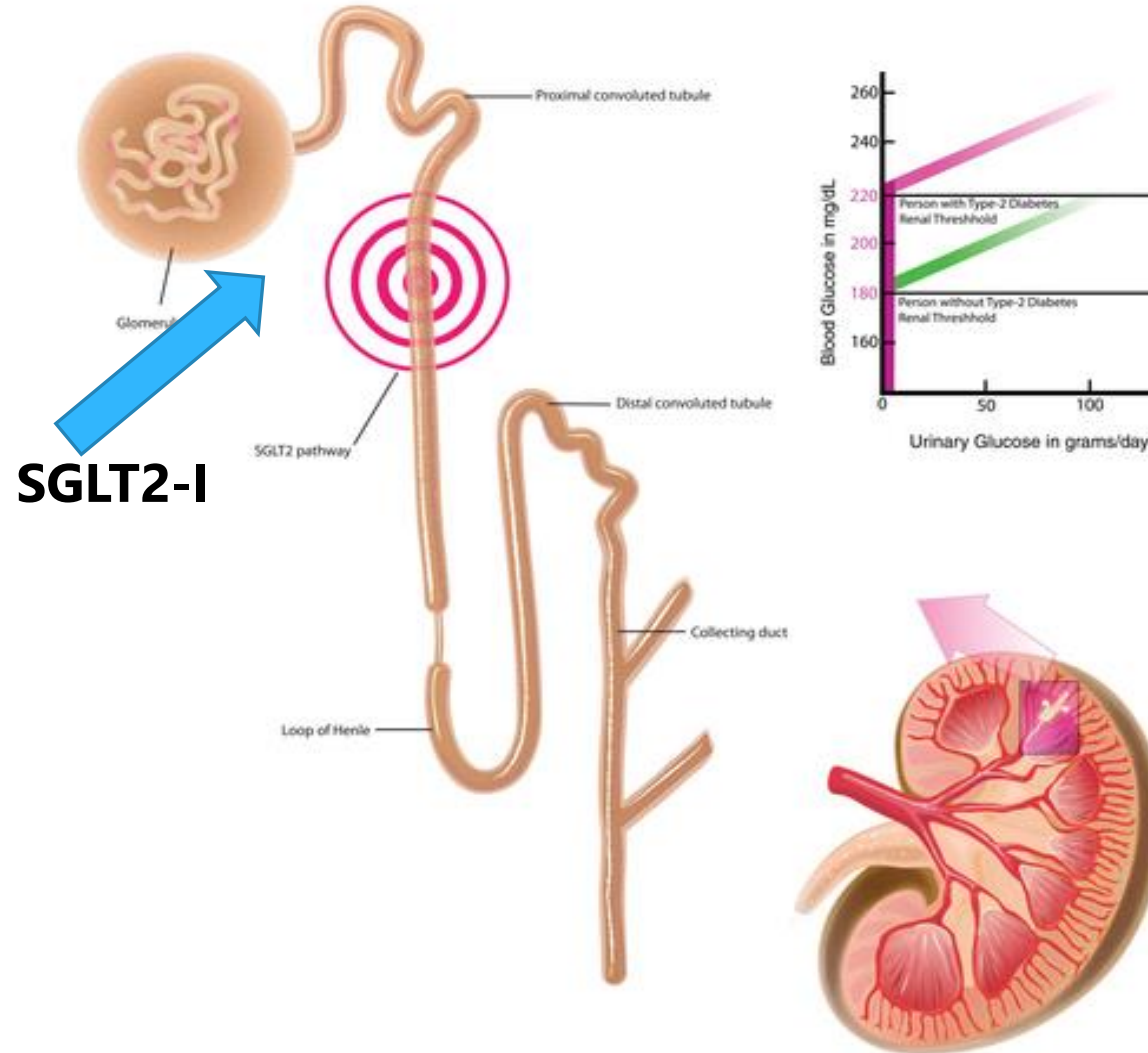
- Cardiorenal risk reduction in addition to healthy lifestyle changes and statins
- Vascular disease (coronary disease, peripheral vascular, stroke)-SGLT2 inhibitors or GLP 1R agonist and GIP agonist therapy
- Heart failure- SGLT2 inhibitor and GLP / GIP R agonists
- Preventing CKD progression- SGLT2 inhibitor (if GFR permissible)

SGLT-2 Inhibitors — Canagliflozin, Dapagliflozin, Empagliflozin

- **Action:**

- Inhibition of SGLT-2 reduces reabsorption of glucose in the proximal tubules
- Blocks approximately 50-80 grams of glucose per day from being reabsorbed
 - Results in increased urinary glucose excretion
 - Decreased plasma glucose levels

SGLT-2 Inhibition

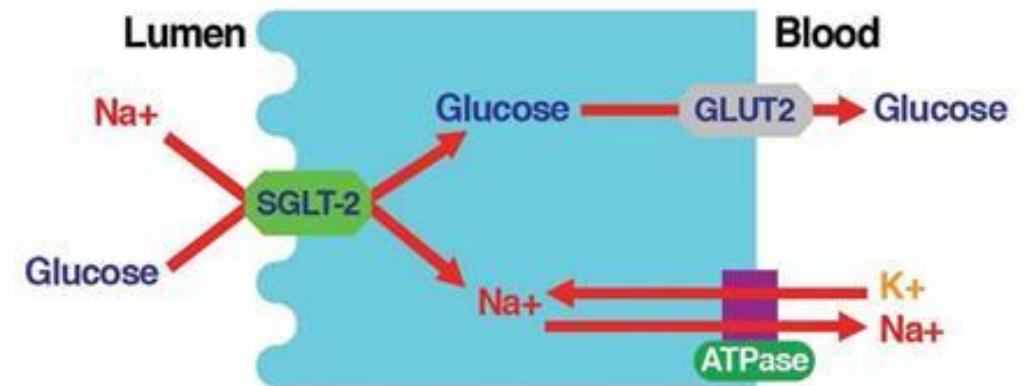
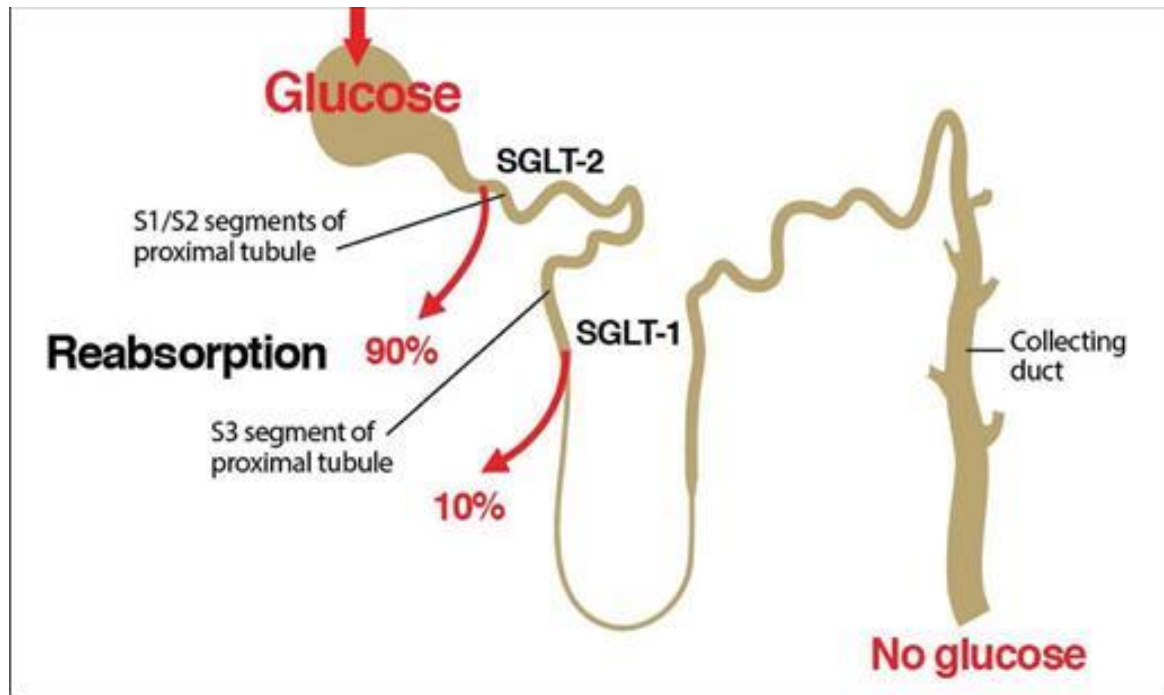


The kidneys of people with type 2 diabetes reabsorb greater amounts of glucose back into the body compared to non-diabetic people, which may contribute to elevated glucose levels. Canagliflozin, an investigational SGLT2 inhibitor, blocks the reabsorption of glucose by the kidney, increasing glucose excretion and lowering blood glucose levels.

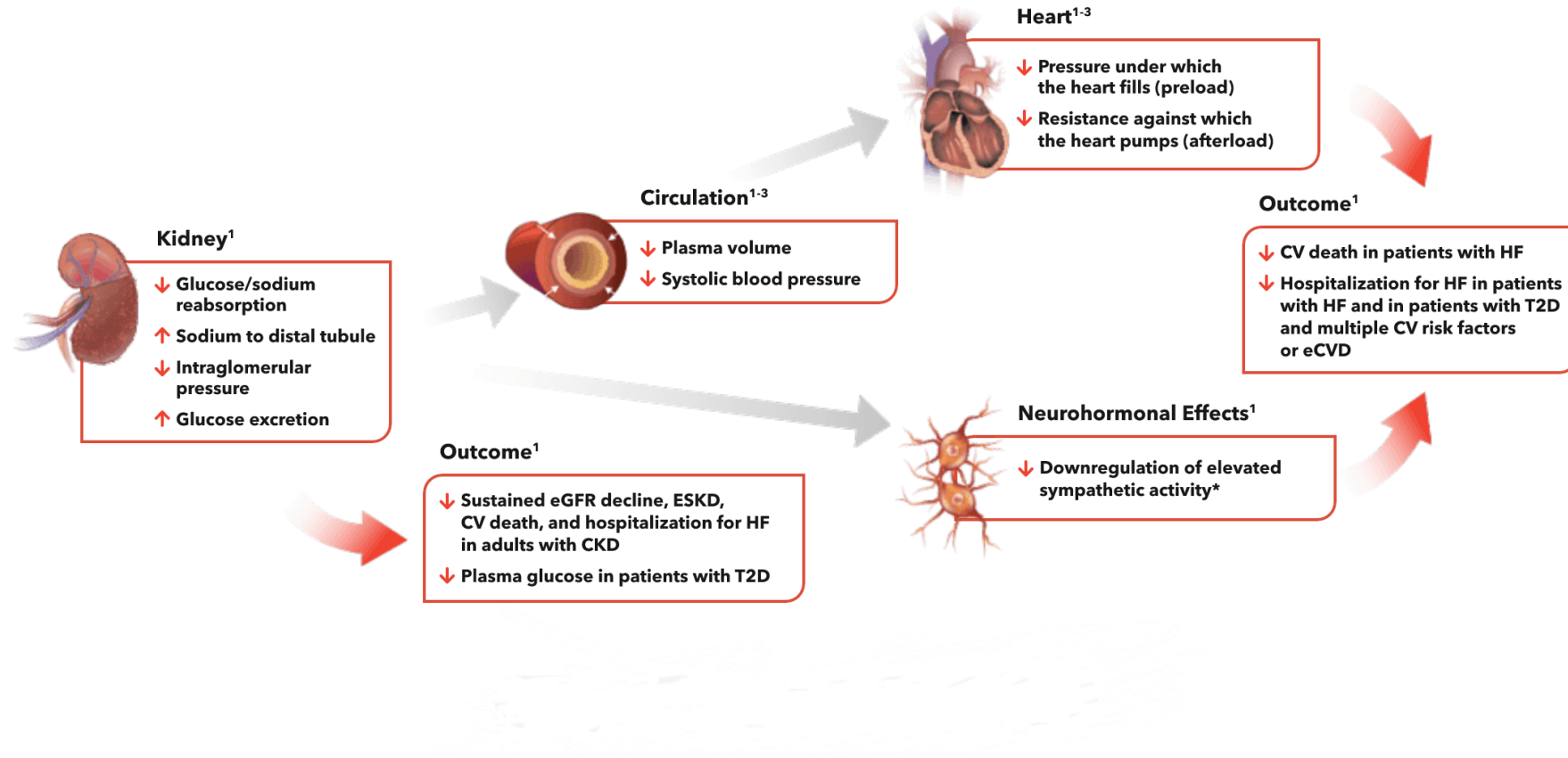
SGLT-2 inhibitors-

glucose excretion can be induced by blocking the activity of the renal sodium-glucose cotransporter 2 (SGLT-2). This mechanism corrects hyperglycemia independently of insulin.

Empagliflozin, Dapagliflozin, Canagliflozin,



SGLT2 inhibitors and Cardiorenal protection



Question 4

The patient begins taking the new medication and tolerates a therapeutic dose without problems. One month later he is admitted to the hospital after a motor vehicle collision in which he sustained a complicated leg fracture. His inpatient treatment is expected to last several days and to include orthopedic surgery with moderate potential for blood loss. How should his diabetes be managed during this admission?

- A. continue all home medications
- B. continue home medications except for metformin
- C. combine A or B with supplemental (“sliding-scale”) short-acting insulin
- D. switch to a basal-bolus insulin regimen using long- and short-acting insulins
- E. switch to supplemental (“sliding-scale”) short-acting insulin alone

Question 4

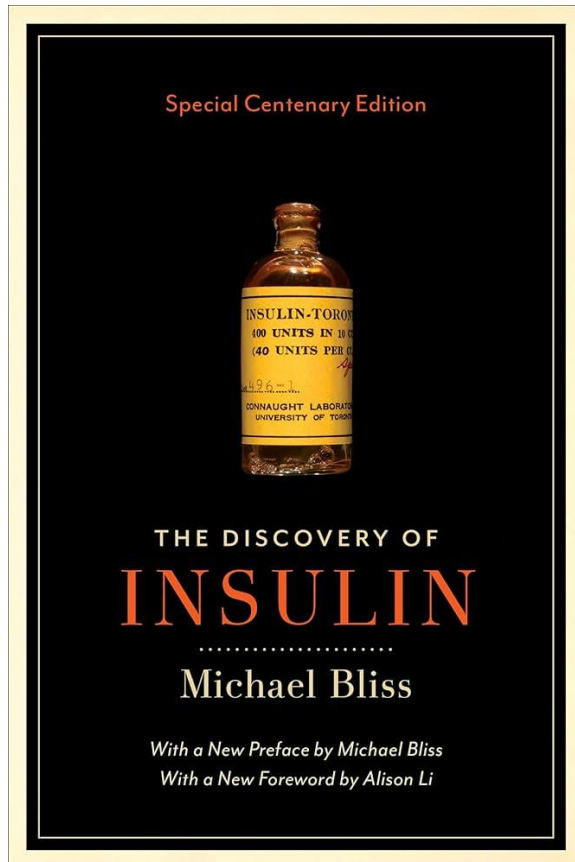
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MALA

- Precipitating factors- Illness, sepsis, dehydration, hypoxia
- DX- high lactic acid(usu over 4) , Ph< 7.35- high AG
- Mechanism – Metformin accumulates disrupting mitochondrial oxidation/ electron transport chain---leads to a shift to anerobic glycolysis (glucose is converted to pyruvate, which is unable to enter the mitochondria).
- Excess pyruvate is converted to lactate--- due to poor perfusion and ARF/ liver ischemia, lactate accumulates.
- Treatment- stop metformin, hydration and treat the cause

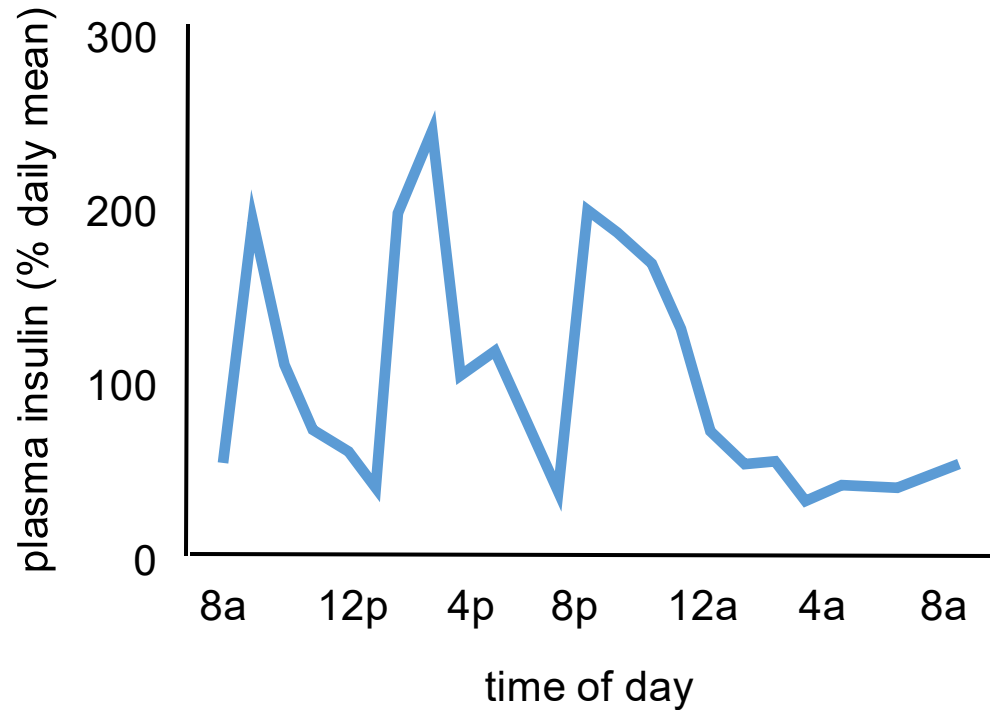
Nobel Prize winner 1923



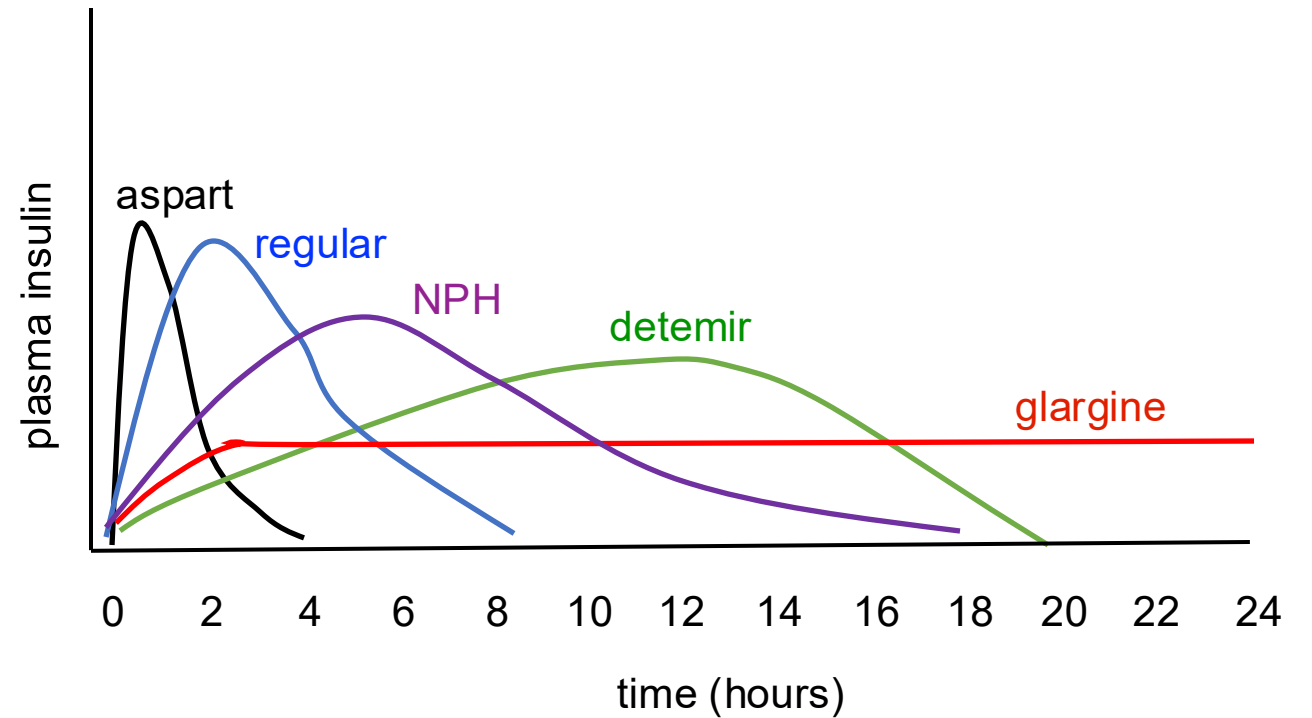
- Short acting – Prandial/ meal
 - Aspart, Lispro, Glulisine etc
 - Inhaled ultra rapid acting insulin
- Long acting – basal
 - Glargine, Detemir, Degludec
 - Once a week insulin Icodec
- Intermediate acting/ combination
 - 70/30 NPH/ regular insulin or 75/25 combo

Insulin therapy

Insulin in healthy individuals



Insulin preparations



Question 5

The patient has an uneventful hospital course, and recovers well from his injuries. He resumes his home/ oral medications , but is unable to bring his sugars under good control. He has gained 12 pounds due to reduced activity. What is the next best step in medication regimen to address above issues?

- A. Intensify physical activity
- B. Add basal insulin back
- C. Continue current therapy
- D. add once a week GLP1 receptor agonist

Question 5

The patient has an uneventful hospital course, and recovers well from his injuries. He resumes his home/ oral medications , but is unable to bring his sugars under good control. He has gained 12 pounds due to reduced activity. What is the next best step in medication regimen to address above issues?

- A. Intensify physical activity
- B. Add basal insulin back
- C. Continue current therapy
- D. add once a week GLP1 receptor or Dual GLP1/GIP agonist

GLUCAGON RELATED PEPTIDES

Intestinal L cells in distal ileum and colon produce different peptides from proglucagon molecule, that include Glucagon like peptides, GLP 1, GLP 2. They are stimulated by a meal and inactivated by protease, Dipeptidyl peptidase 4 (DPP-4)

GLP1-

Pancreas- stimulates insulin secretion, inhibits glucagon secretion

Stomach -Inhibits Gastric emptying

Brain-Inhibits appetite

GLP2- inhibits intestinal motility

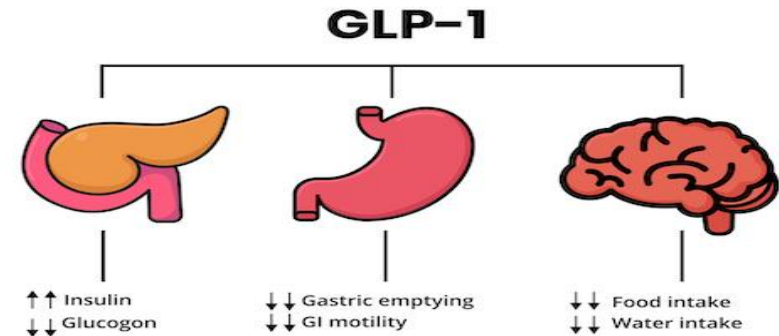
GIP--Produced by K cells in duodenum and jejunum, in response to glucose.

Pancreas -Stimulates Insulin, glucagon and somatostatin secretion.

Stomach-Inhibits gastric emptying.

Fat-Stimulates adipogenesis

GLP –1 agonist



GLP 1 agonist- Dulaglutide, Semaglutide, Liraglutide
GLP1+ GIP agonist – Tirzepatide
oral pills – Orforglipron, Rybelsus

- **GLP-1 Ras/ GIP RA stimulate the body's own insulin secretion**
- Glucagon-like peptide-1 (GLP-1) and GIP are a physiological hormones that has multiple actions on glucose.
- Enhanced appropriate pancreatic β -cell (insulin and amylin) secretion
- Pancreatic α -cell (glucagon) suppression
- Decreased liver glucose production
- Increased satiety through central nervous system
- Slowed gastric emptying time
- Increased insulin uptake in peripheral tissue via weight loss

Question 6

The patient goes on a cruise and forgets to take his medications. As the sugars were running high, the ship's doctor (You) start him on glipizide 10 mg bid. The sugars are better. Patient contracts a viral gastroenteritis 2 days later and his wife finds him incoherent, with clammy sweaty skin. What is the most likely cause of his symptoms and what will you do next?

- A. Hypoglycemia and stat blood glucose
- B. viral illness and you get CBC, BMP
- C. Viral illness and get chest Xray

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- Glucose is 35

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- What did you do as an emergency treatment?
 1. Orange juice
 2. Injection of glucagon
 3. Steroid injection
 4. Antibiotics injection

- What did you do as an emergency treatment?

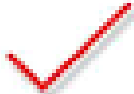
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For treatment of hypoglycemia

- Level 1 Glucose <70 mg/dL and ≥ 54 mg/dL - oral glucose
- Level 2 Glucose <54 mg/dL
- Level 3 A severe event characterized by altered mental and/or physical status requiring assistance for treatment of hypoglycemia, irrespective of glucose level

Clinical/biological risk factors	Social, cultural, and economic risk factors
Major risk factors • Recent (within the past 3–6 months) level 2 or 3 hypoglycemia • Intensive insulin therapy* • Impaired hypoglycemia awareness • End-stage kidney disease • Cognitive impairment or dementia	Major risk factors • Food insecurity • Low-income status§ • Homelessness • Fasting for religious or cultural reasons
Other risk factors • Multiple recent episodes of level 1 hypoglycemia • Basal insulin therapy* • Age ≥75 years† • Female sex • High glycemic variability‡ • Polypharmacy • Cardiovascular disease • Chronic kidney disease (eGFR <60 mL/min/1.73 m² or albuminuria) • Neuropathy • Retinopathy • Major depressive disorder	Other risk factors • Low health literacy • Alcohol or substance use disorder

How to Treat Low Blood Sugar (Hypoglycemia)

1.  Eat/Drink 15 g Carbs
2.  Wait 15 Minutes
3.  Check Blood
4.  Less than 70 mg/dl?
Repeat Steps 1-4



Wrap-up

- Pharmacotherapy for type 2 diabetes begins with metformin, but doesn't end there.
- After metformin, —safety, tolerability, efficacy, price, and simplicity can all influence the choice.
- Safety and dosing can be affected by pharmacokinetic and pharmacodynamic interactions, and by other medical problems.
- Patients with type 2 diabetes often have other medical issues that call for attention and can affect treatment.

Questions?